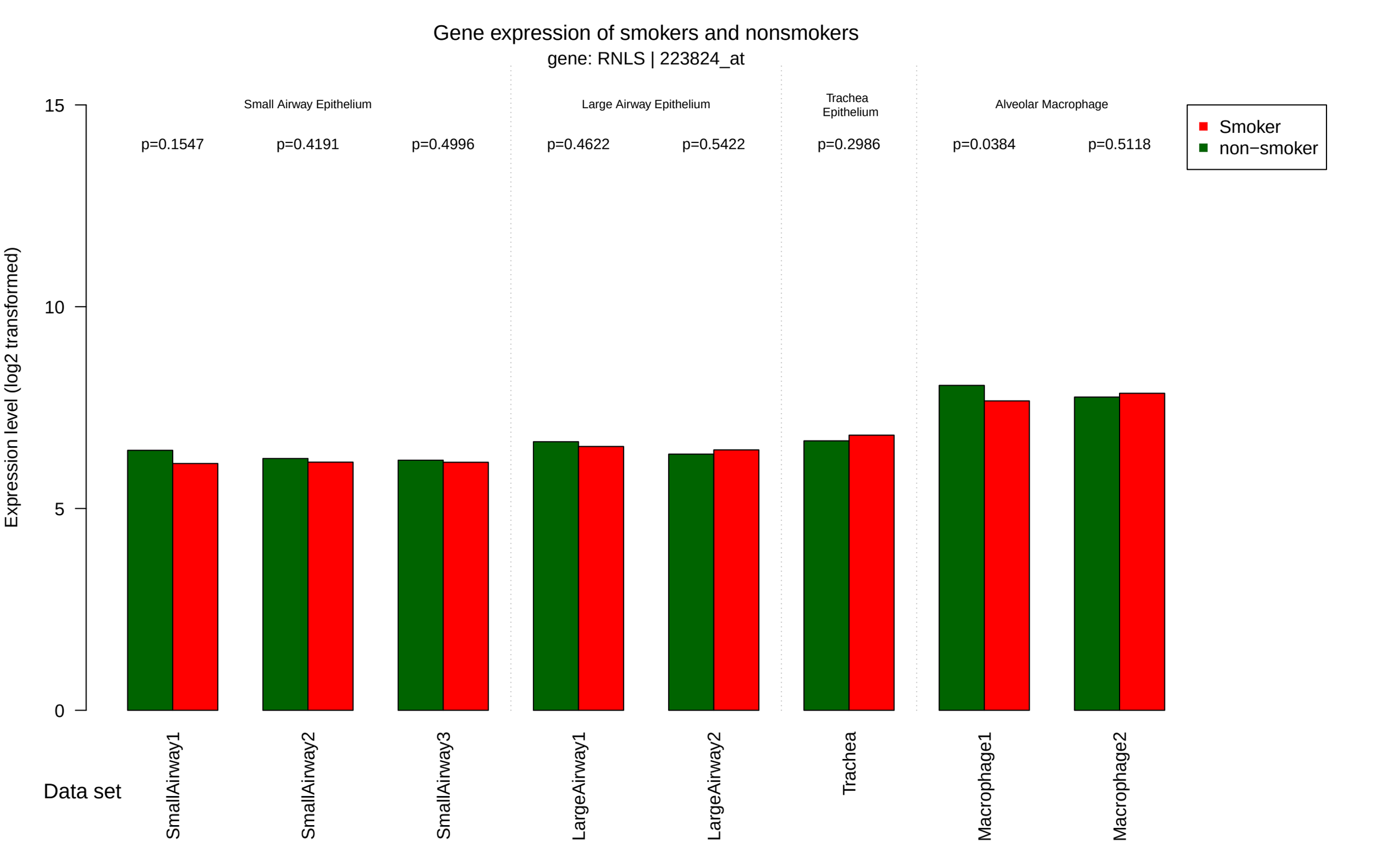
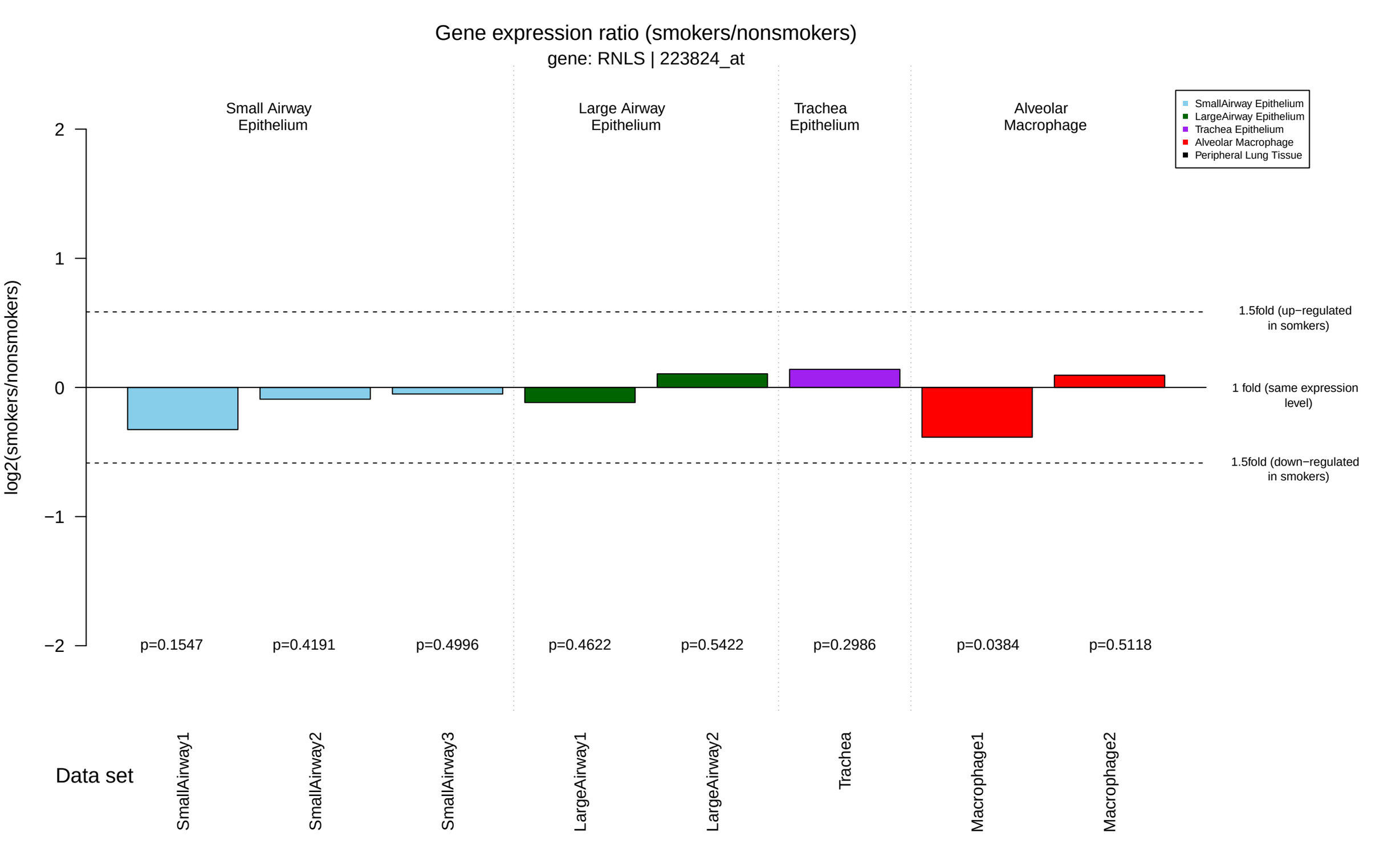
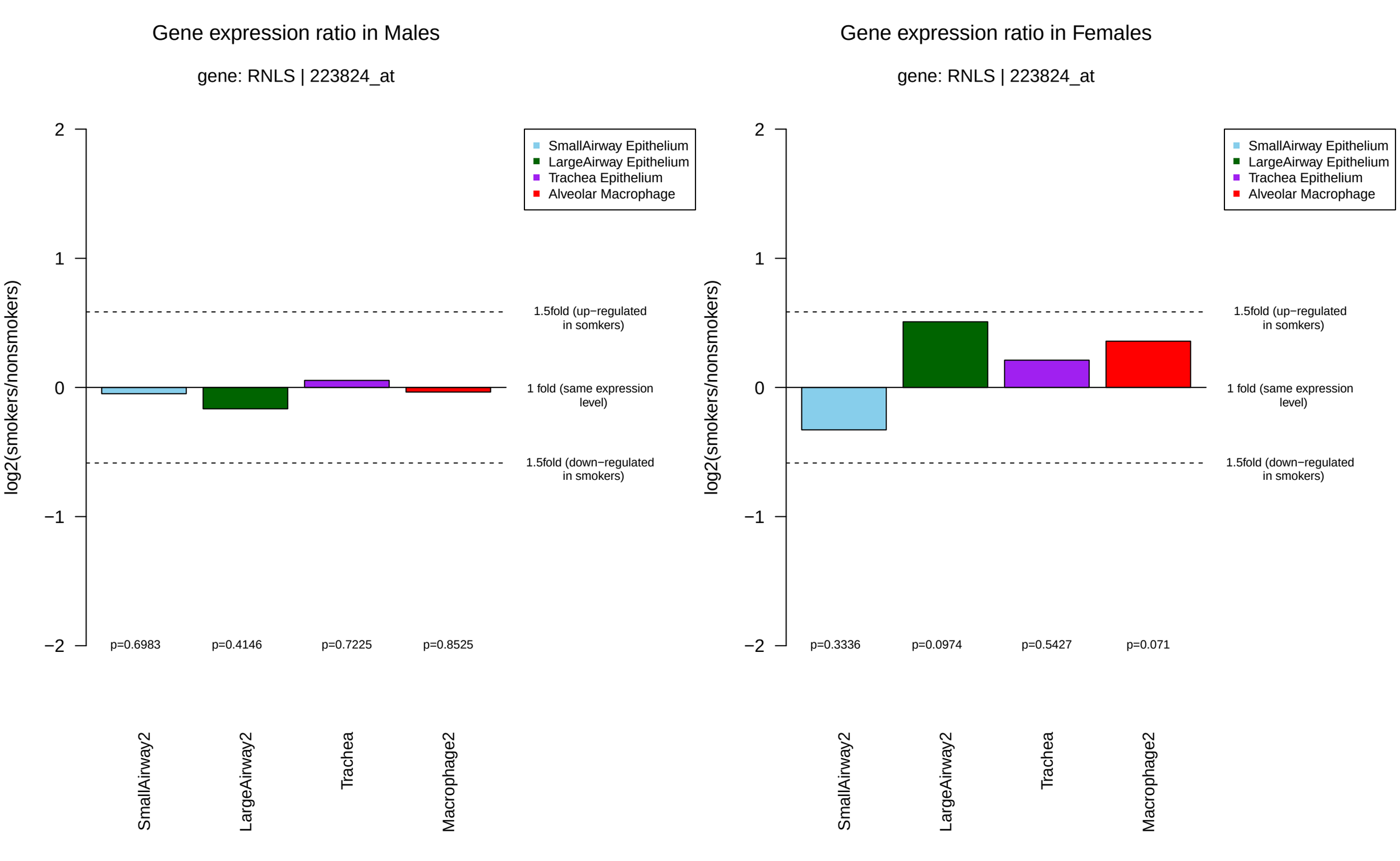
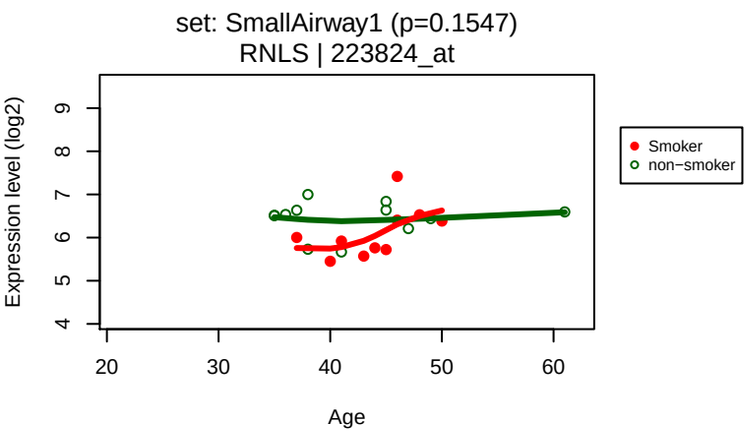
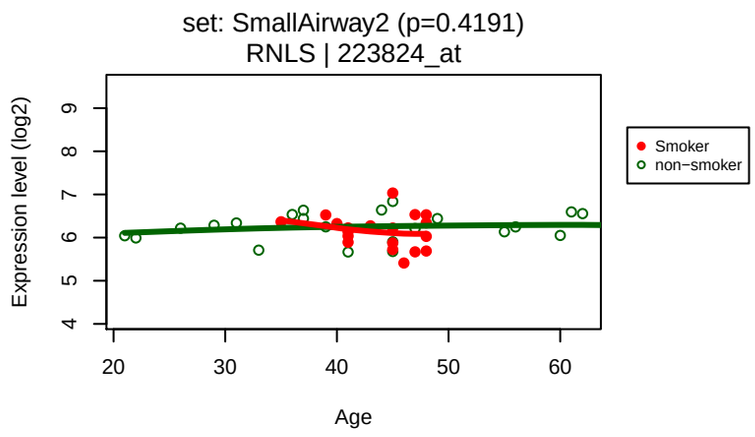
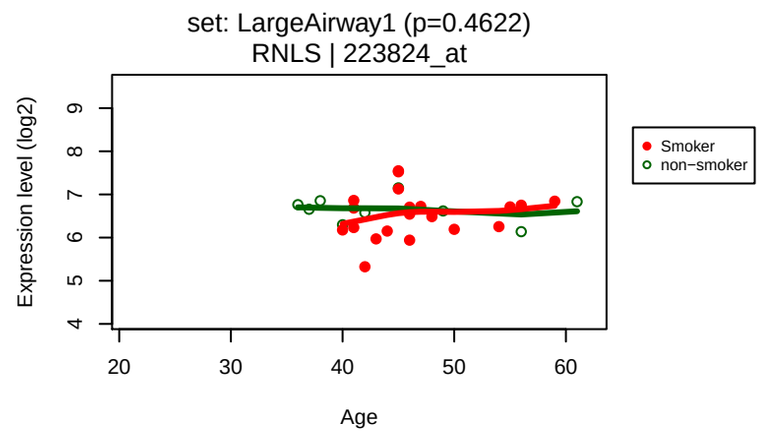

Gene expression of smokers and nonsmokers
gene: RNLS | 223824_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1547
p=0.4191
p=0.4996
p=0.4622
p=0.5422
p=0.2986
p=0.0384
p=0.5118
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

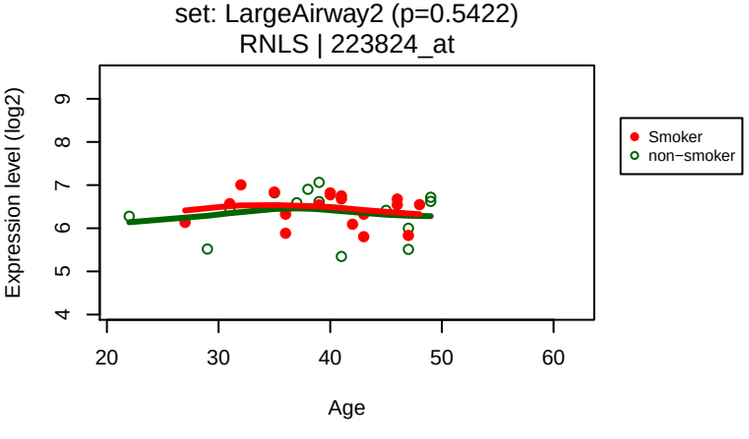

Gene expression ratio (smokers/nonsmokers)
gene: RNLS | 223824_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1547
p=0.4191
p=0.4996
p=0.4622
p=0.5422
p=0.2986
p=0.0384
p=0.5118
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

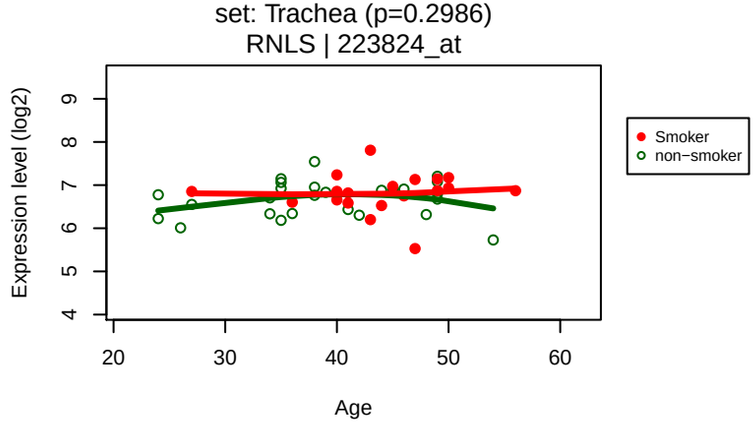

Gene expression ratio in Males
gene: RNLS | 223824_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6983
p=0.4146
p=0.7225
p=0.8525
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RNLS | 223824_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3336
p=0.0974
p=0.5427
p=0.071
SmallAirway2
LargeAirway2
Trachea
Macrophage2

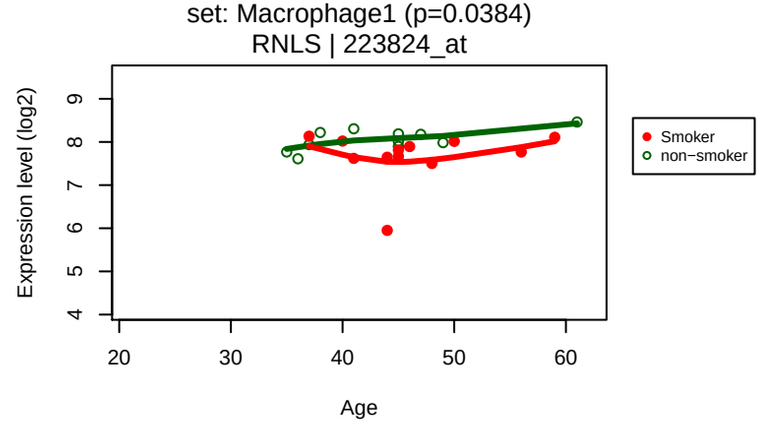

set: SmallAirway1 (p=0.1547)
RNLS | 223824_at
Expression level (log2)
Age
Smoker
non−smoker

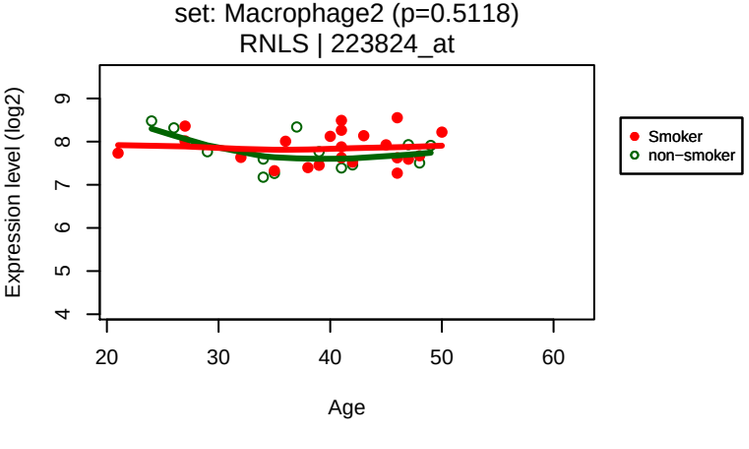

set: SmallAirway2 (p=0.4191)
RNLS | 223824_at
Expression level (log2)
Age
Smoker
non−smoker

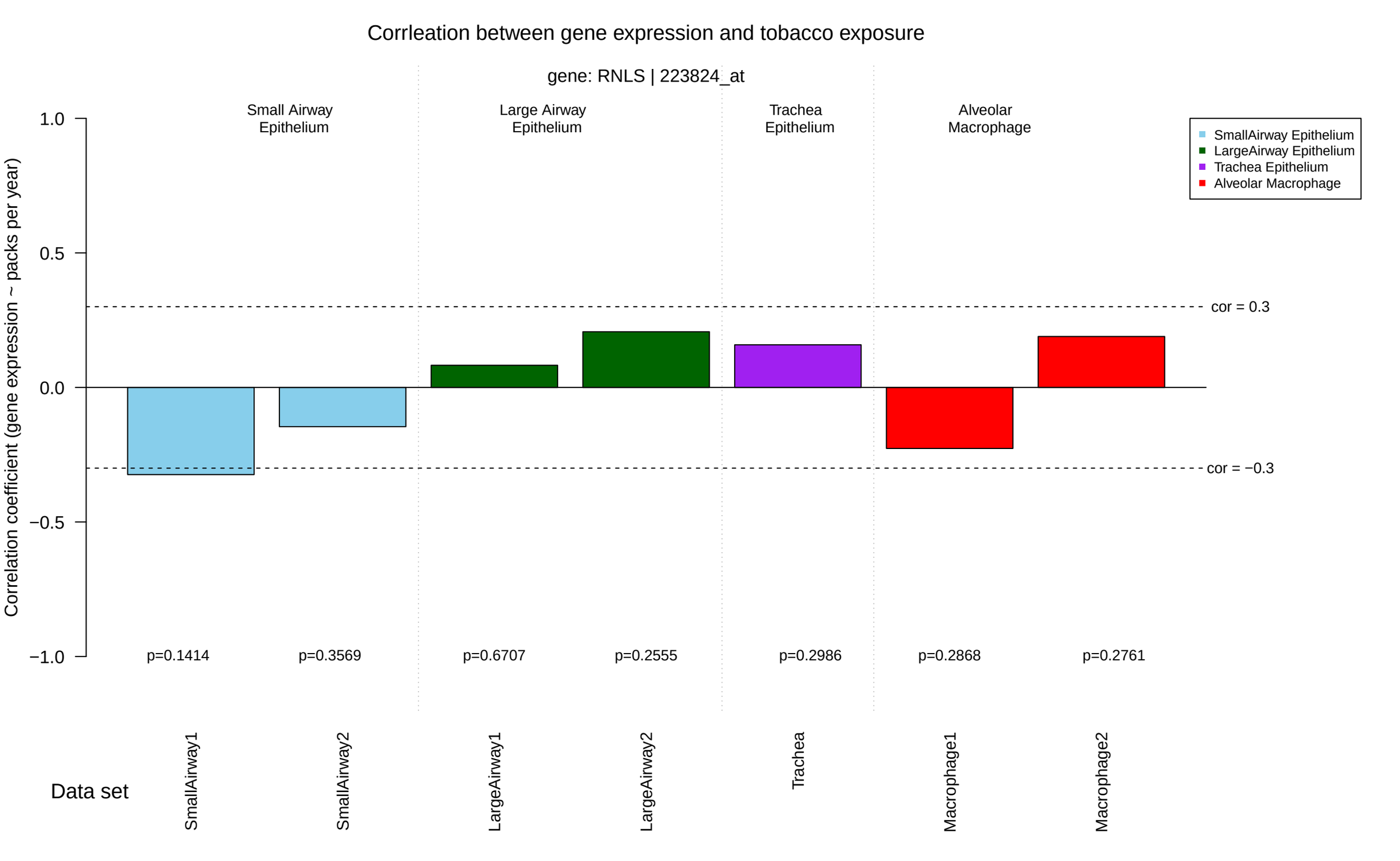

set: LargeAirway1 (p=0.4622)
RNLS | 223824_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5422)
RNLS | 223824_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2986)
RNLS | 223824_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0384)
RNLS | 223824_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5118)
RNLS | 223824_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RNLS | 223824_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1414
p=0.3569
p=0.6707
p=0.2555
p=0.2986
p=0.2868
p=0.2761
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2